THE SPACE OF A LOCALLY NOETHERIAN ALGEBRAIC STACK IS LOCALLY NOETHERIAN, AND NOETHERIAN WHEN THE STACK IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement for schemes that the space of a locally Noetherian scheme is a locally Noetherian topological space. Invoked inside the counted proof (source lines 2266-2268) as a cross-chapter pointer. Provides the property for the presenting scheme.
- The statement that the image of a locally Noetherian topological space under an open surjective map is locally Noetherian. Invoked inside the counted proof (source lines 2269-2272) as a cross-chapter pointer. Descends local Noetherianity from the presentation to the space of the stack.
- The statement that a quasi-compact locally Noetherian topological space is Noetherian. Invoked at the end of the counted proof (source lines 2274-2276) as a cross-chapter pointer. Gives the second assertion once quasi-compactness is known.

Source reference. Original reference properties-lemma-Noetherian-topology: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2268–2268. Navigation pointer only: the counted text refers to the Stacks item properties-lemma-Noetherian-topology, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference topology-lemma-image-Noetherian: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2272–2272. Navigation pointer only: the counted text refers to the Stacks item topology-lemma-image-Noetherian, which belongs to another chapter of the project and is not part

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Source reference. Original reference topology-lemma-quasi-compact-locally-Noetherian-Noetherian: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2276–2276. Navigation pointer only: the counted text refers to the Stacks item topology-lemma-quasi-compact-locally-Noetherian-Noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack.*

(1) *If $\mathcal{X}$ is locally Noetherian then $|\mathcal{X}|$ is a locally Noetherian topological space.*
(2) *If $\mathcal{X}$ is quasi-compact and locally Noetherian, then $|\mathcal{X}|$ is a Noetherian topological space.*

Proof. Assume $\mathcal{X}$ is locally Noetherian. Choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. As $\mathcal{X}$ is locally Noetherian we see that U is locally Noetherian. By Properties, Lemma properties-lemma-Noetherian-topology this means that $|U|$ is a locally Noetherian topological space. Since $|U| \to |\mathcal{X}|$ is open and surjective we conclude that $|\mathcal{X}|$ is locally Noetherian by Topology, Lemma topology-lemma-image-Noetherian. This proves (1). If $\mathcal{X}$ is quasi-compact and locally Noetherian, then $|\mathcal{X}|$ is quasi-compact and locally Noetherian. Hence $|\mathcal{X}|$ is Noetherian by Topology, Lemma topology-lemma-quasi-compact-locally-Noetherian-Noetherian. □